MONASH University

MALAYSIA

FIT5226 Multi-agent System & Collective Behaviour

Unit Synopsis

Fuad Noman

School Of Information Technology

Monash University Malaysia

What's this course about?

1. The unit **is** about real-world problem-solving with multi-agent system (MAS).
2. The unit **is** about modeling and analyzing decision-making in the context of multiple agents using reinforcement learning and game theory.
3. The subject **is not** (mainly) about programming.
4. The subject make use of Python as the programming language in which lab work (etc.) is done.

Broad overview of the unit's weekly schedule

Week 1-2

- Introduction and scoping & macroscopic systems (deterministic)
- Macroscopic Systems (stochastic) & RL and MDPs Introductions

Week 3-4

- Bellman Equations & Policy and Value Iteration
- SARSA & Q-Learning

Week 5

- Deep-Q

Week 6

- Game Theory Introduction

Broad overview of the unit's weekly schedule

Week 7-8

- Solution concepts; Nash equilibrium & Markov games, MiniMax-Q
- Population Games and Evolutionary Game Theory

Week 9

- Social Dilemmas

Week 10

- Introduction to Coalition Game Theory

Week 11

- Voting

Week 12

- Auctions

Course material

- Your main portal will be, as you already know, the unit's **Moodle** page.
- Material available on Moodle will include:
  - Lecture notes and slides
  - Additional references
  - Practical (Lab) sheets
  - Tutorial sheets
  - Lecture recordings (MUA)

- The unit's **Ed discussion** page will be used to provide all unit announcements and is the best place to post queries about the unit.

Main Textbook References

1. Shoham, Yoav, and Kevin Leyton-Brown. **Multiagent systems: Algorithmic, game-theoretic, and logical foundations**. Cambridge University Press, 2008
2. Leyton-Brown, Kevin, and Yoav Shoham. **Essentials of game theory: A concise multidisciplinary introduction**. 2008).
3. Richard S. Sutton and Andrew G. Barto, **Reinforcement Learning: An Introduction**.2018.
4. Wooldridge, Michael. **An introduction to multiagent systems**. 2009.

This subject will aid your development into professional programmers, technicians, computer scientists, etc. Of course, you are strongly encouraged to build on this and read beyond what is prescribed for this unit

Course Structure

- **Lecture**: 2 hours/week Workshop (refer to timetable for time/venue)
- **Tutorial+Lab**: 2 hours weekly(refer to timetable for time/venue)
- **Assessments**: 4 assessments,
  - Assessment 1 – Project Stage 1 – 10% – due end of week 5.
  - Assessment 2 – Project Stage 2 – 10% – due end of week 8.
  - Assessment 3 – Quiz – 25% – week 9.
  - Assessment 4 – Project Stage 3 – 55% – due end of week 12.

Marks and Hurdle -- IMPORTANT

To pass FIT5226

✶ Your marks must average to **at least** 50% of the total marks for this unit.

- In-Semester Assessment contributes to 100% of total unit marks
  - Assessed marks = $\frac{100}{100}$ (across 4 assessments)

Student Responsibilities

Responsibilities

- ✶ Regularly attend the weekly activities.
- ✶ Diligent self-study (nominally 8 hours/week).
- ✶ This subject is best understood practically. Internalizing its concepts by coding them up is the *key* to excel in it.

Etiquette

- ✶ No noise and distractions during the Tutorial/Lab.
- ✶ Mute your microphone if you are not currently using it. Be polite and patient.

Plagiarism and Cheating

- Monash University takes plagiarism and cheating very seriously. There are severe penalties for them.
- Read Monash University's [Plagiarism and cheating Policy]
- It is **OKAY** to work together in trying to understand concepts. Peer Assisted learning (PAL) is fun and a great way to **cross-fertilize** each others' thinking – but each student must be conscientious in her/his work, write code for the entire assignments alone, and be able to explain and modify it on request
- All scripts will be screened through a powerful software for detecting software plagiarism.

SEE YOU NEXT WEEK